adyen CAREERS

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Machine Learning Scientist

Development Amsterdam

This is Adyen

Adyen provides payments, data, and financial products in a single solution for customers like Meta, Uber, H&M, and Microsoft - making us the financial technology platform of choice. At Adyen, everything we do is engineered for ambition.

For our teams, we create an environment with opportunities for our people to succeed, backed by the culture and support to ensure they are enabled to truly own their careers. We are motivated individuals who tackle unique technical challenges at scale and solve them as a team. Together, we deliver innovative and ethical solutions that help businesses achieve their ambitions faster.

Machine Learning Scientist

Adyen is looking for a Machine Learning Scientist to join our team in Amsterdam, a person sitting at the cornerstone of algorithms, mathematics, and engineering, who can solve problems by designing and implementing production-ready machine learning solutions. You will be responsible for building, developing and deploying algorithms that power data products at Adyen.

We are currently hiring for the following teams:

Insights - Diagnostics: The Insights team is at the core of this platform, providing the world's largest merchants with the data and analytics they need to optimize their payment performance. Within this, our Proactive Diagnostics initiative acts as a proactive guard for merchant revenue, closing the loop between detecting an anomaly and

providing a clear path to rectification. We operate at the intersection of Big Data and actionable intelligence. By leveraging Adyen's global payment flow, we apply advanced statistical models and Causal Inference to not only detect performance drops but to explain the "why" behind them. We are looking for a Machine Learning Engineer to help us architect the next generation of our diagnostic engine.

Regulatory Reporting Technology:

Adyen's Regulatory Reporting Tech team is seeking a Machine Learning Scientist to join us in Amsterdam. You will help in further automating and scaling our global regulatory reporting framework to keep Adyen compliant across all markets. Together with the team, you will implement key technical solutions to streamline our regulatory reporting operations. If you have a strong machine learning background and love solving complex problems, we want to hear from you.

In this role, you will:

- Research, design, implement, train, deploy and monitor machine learning algorithms, either for batch prediction or real-time. Examples include: on-line learning algorithms to pick the best optimization decision in a changing environment, clustering algorithms to group customers/shoppers, supervised and semi-supervised learning methods for inference on risk patterns or graph analysis, representation learning for behavior prediction and monitoring, Anti-Money Laundering (AML) systems and real-time anomaly detection based on time-series modeling;
- Develop orchestrated pipelines for analytical purposes and machine learning training;
- Contribute to ongoing automation efforts for our experiments, training runs, validation runs and monitoring before, during and after deployment. Furthermore, collaborate with MLOps to improve our machine learning tooling;
- Collaborate closely with product managers and business stakeholders to understand requirements, define problems, and frame them as solvable machine learning tasks;
- Explore and analyze large, complex datasets to identify patterns, insights, and opportunities for ML-driven solutions;
- Define key performance metrics, design rigorous experiments (e.g., A/B tests), and perform statistical analysis to validate model performance and quantify business impact;

Who You Are:

- You have 3+ years of experience as a machine learning or data scientist;
- You have experience with the full machine learning model lifecycle in production flows;

- You have experience leveraging a big data framework to create the pipelines needed to feed the models with appropriate data;
- You have a good understanding of software engineering practices as well as data engineering and MLOps principles;
- You have knowledge of data science and statistics and machine learning techniques. These include a strong grounding in statistical inference, machine learning for prediction, and causal inference - Deep Learning experience is a plus;
- You have strong familiarity with the standard data science toolkit, such as (py)spark, (Trino) SQL, Tensorflow, PyTorch, XGBoost/LightGBM, Pandas, MLFlow or similar MLOps frameworks, and Airflow;
- You have an experimental mindset with a launch fast and iterate mentality. Extensive experience with designing and running experiments is essential;
- You are proactively taking the lead in projects, from ideation to deployment. You have experience working with a wide range of stakeholders and can clearly communicate complex outcomes over a wide range of audiences.
- You stay up-to-date with the latest research and developments in the machine learning field, and apply this where valuable.

Our Diversity, Equity and Inclusion commitments

Our unique approach is a product of our diverse perspectives. This diversity of backgrounds and cultures is essential in helping us maintain our momentum. Our business and technical challenges are unique, and we need as many different voices as possible to join us in solving them - voices like yours. No matter who you are or where you're from, we welcome you to be your true self at Adyen.

Studies show that women and members of underrepresented communities apply for jobs only if they meet 100% of the qualifications. Does this sound like you? If so, Adyen encourages you to reconsider and apply. We look forward to your application!

What's next?

Ensuring a smooth and enjoyable candidate experience is critical for us. We aim to get back to you regarding your application within 5 business days. Our interview process tends to take about 4 weeks to complete, but may fluctuate depending on the role. Learn more about our hiring process here. Don't be afraid to let us know if you need more flexibility.

This role is based out of our Amsterdam office. We are an office-first company and value

in-person collaboration; we do not offer remote-only roles.

Apply now

The Adyen Formula

The way we work is guided by the eight principles of the Adyen Formula. Learn more here.

Learn more >

Tech careers

Our engineers are building the first financial technology platform that combines payments, data, and financial services. We're looking for more talented problem solvers to help us address a unique set of challenges.

Learn more >

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